

Prompt-ReID: Prompt Based Inference for Robust Re-Identification

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Abstract

Large-scale pre-trained Vision-Language Models (VLMs) such as CLIP have demonstrated superior performance and robustness for various computer vision tasks. Recent works that utilize CLIP have shown that it is possible to fine-tune its image encoder while preserving its ability to generalize to a given downstream task. However VLMs also contain a text encoder along with usual image encoder, which is often underutilised for downstream tasks, especially in re-identification. Recent methods in re-identification only utilize the conventional approach of nearest image search in the image encoder’s representation space at inference. Hence, in this work we aim to utilize the unused text encoder to generate an identity specific prompt in the gallery and perform a similarity search between the prompt embeddings for each identity and the image features for a given query image. We demonstrate that the prompt-based inference significantly improves not just the accuracy on the in-distribution evaluation set but also the out-of-distribution (OOD) generalization performance across various re-identification benchmarks.

1. Introduction

Image Re-Identification (ReID) involves matching a query image of an object across different, often non-overlapping, camera views in a surveillance system. The goal is to determine if the object has appeared in another location at a different time, captured by the same or a different camera [15, 27, 43, 46]. State-of-the-art (SotA) approaches focus on learning a robust image encoder to produce a unique feature representation for each object, enabling effective matching across diverse conditions. The challenges

involved include variations in background clutter, occlusion, pose, illumination, and image resolution [47, 66, 77].

Early research on deep learning-based ReID methods primarily relied on Convolutional Neural Networks (CNNs) [49, 50, 53]. A popular strategy for this involves learning a metric space using loss functions like triplet loss [13] and an additional identity loss function [73], similar to the cross-entropy loss used in multi-class classification. However, CNNs only focus on analyzing smaller regions in the image [38], that hinders their ability to capture the global appearance crucial for effective ReID. Vision Transformers (ViTs) [9] address this limitation by inherently attending to all parts of the image simultaneously, this coupled with their demonstrated superior performance in recent works [11, 77, 78], makes them a the preferred choice for ReID tasks. However, ViTs are generally more parameterized than CNNs, necessitating a careful integration with pre-training strategies for optimal utilization. Vision Language Models (VLMs) based on ViTs like CLIP [42], pre-train their image and text encoders on a large image-caption dataset to mitigate overparameterization issues, resulting in image and text encoders that produce representations rich in semantic information. The contrastive pretraining of CLIP results in a joint text and image representation space. For downstream tasks like image classification, a common approach for VLMs is to use text prompts. These are essentially text templates customizable for each category. They are used to perform a similarity search between a given image and the set of prompts [2, 5, 74, 75]. However, prompt-based techniques for inference are unexplored in ReID tasks we explore them in this work.

ReID involves assigning the correct identity for a set of unknown images called a query set. This is done by comparing it against a set of known images called a gallery set.

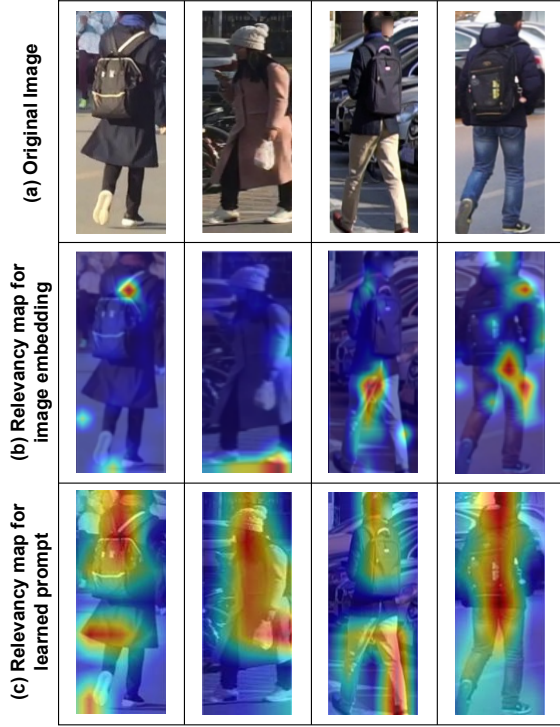


Figure 1. We analyze the relevance map from an image encoder fine-tuned with CLIP-ReID on the MSMT17 dataset (a). We use [3] where the [CLS] token of one modality is used to compute the relevance map of the other modality. For each image in (b), we use the image embedding of a gallery image with the same identity as the given query image and calculate the relevancy map using the cross attention of [CLS] token embedding of the former with the patch tokens of the latter. This is compared to the map generated with learned prompt tokens (**CLIP-ReID**) (c). Learned prompt embeddings capture major parts of the person, while image encoder embeddings highlight localized details. (**Ours**)

Conventionally, this is done by finding the closest image (nearest neighbor search) in the gallery set to a given query image, similar to the image-retrieval task. This search happens in a representation space defined by the image encoder, and the gallery images are ranked based on a pre-defined distance measure, typically the Euclidean (L_2) distance. Performance metrics like Rank-K (RK) measure the proportion of queries for which the correct match is among the top K candidates from the gallery retrieved by the model. Another commonly used metric is mAP, which denotes the average precision (AP) computed at rank K across all query images present in the set.

Recently, a few methods in ReID utilize VLMs [6, 16, 22, 32, 65]. However, they do not make use of the generalization capability of the text encoder [14], as at inference

only the image encoder is used to get the nearest neighbor image in the representation space. Such methods are fragile as individual image’s representation can be perturbed with occlusions leading to incorrect results. To mitigate this, we propose using text encoders to learn identity-specific prompts. In this approach, we encode the semantics common for all images for a given identity in a text prompt, which is more robust and reliable (Fig. 1). This is especially useful where we require the learned text representations to be robust against distribution shift [28–30]. This problem aligns with the task of out of domain generalization, where an image encoder fine-tuned on a source dataset needs to perform well on an unseen target dataset.

Our work leverages the superior performance of VLMs trained on a specific type of similarity measure (cosine-similarity) for prompt-based inference in ReID. Our contributions are summarised as follows:

1. We propose a novel inference strategy for ReID that uses the text encoder by creating a set of unique learned prompts for each identity. Our work shows the effectiveness of the nearest prompt search over the conventional nearest image search during inference.
2. We demonstrate the robustness of our inference method by extensively testing our method Prompt-ReID on various datasets and demonstrate its effectiveness in improving performance on both the standard and domain generalization benchmarks for ReID.
3. We show that prompt embeddings capture the semantics of identity and are more robust than individual image embeddings.

2. Related Work

Image ReID is industrially relevant in surveillance systems [18, 39, 51], and hence, recent works have predominantly focused on person and vehicle ReID. The SotA approaches [40, 55, 67, 68] are deep learning based with a loss function to learn a metric space. Metric learning-based losses [45, 48, 60, 69, 77] focus on learning an image representation space defined by an image encoder. The objective is to learn an image encoder that generates a unique image feature vector for all images of the same identity, regardless of variations in illumination, pose, occlusion, and image quality. Some common loss functions include the triplet loss [13, 21] that minimizes the distance between images belonging to the same identity while maximizing the distance between images of different identities in the representation space. Some other variants include quadruplet loss [4] and cosine-margin loss [8]. Some recent works use human parsing to build part-based features which are better able to understand human structure [26]. Often, mining [62, 64] is done to look for hard samples that include the

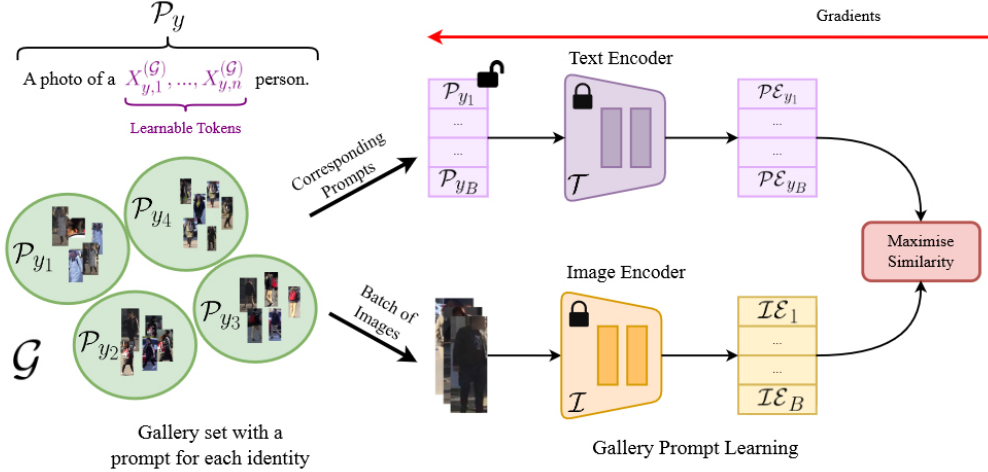


Figure 2. Overview of prompt learning for inference on a gallery set. Each identity $y \in \mathcal{Y}_G$ has a specific prompt $\mathcal{P}_y$ with learnable tokens that encode its semantics, learnt by maximizing similarity.

furthest positive sample and the nearest negative sample to compute these metric losses.

Vision-Language Models such as CLIP [42] integrate text and image representations in a shared embedding space, allowing encoders to capture nuanced semantic information. Combining over-parameterized architectures, like ViTs, with the contrastive loss pre-training of CLIP results in substantial performance improvements. CLIP employs a two-directional InfoNCE loss [37] for aligning images and text. CLIP’s image and text encoders are widely used in downstream tasks like object detection [52], semantic segmentation [59], and image retrieval [1].

Context-Optimization (CoOp) [75] stands as a seminal work in learning text prompts for vision language models. The paper proposes two strategies for learning prompts for image classification. The first is context-specific tokens that do away with hand-engineering prompts for a given context. The second **class-specific tokens** further encodes class-specific information. The image and text encoders are frozen, while only specific prompt tokens are kept learnable. CoOp not only shows superior performance in image classification tasks with a pre-trained CLIP backbone but also the ability to generalize well under the few-shot setting. Despite having very few learnable tokens (4), they can capture both context-specific and class-specific information. Further works [19, 74] have shown that it is possible to learn prompts that contain not just semantic information through the text encoder but also the image encoder.

CLIP-ReID [23] is one of the first works that could successfully leverage a Vision-Language model, for ReID tasks. Their idea revolves around learning a suitable prompt for each identity and using them as an anchor to tighten the cluster of all the corresponding images around it. CLIP-

ReID uses a two-stage pipeline to achieve this. Stage 1 starts with a learnable prompt template and employs the CoOp approach to get to the optimal prompt for each identity. Like CLIP’s training, they utilize the bidirectional InfoNCE losses $\mathcal{L}_{i2t}$ and $\mathcal{L}_{t2i}$ to optimize the prompts. Stage 2 utilizes the text embeddings of these learned prompts as cluster centers and optimizes the image encoder to bring the image embeddings closer to their respective prompt embeddings. They use the standard Identity loss $\mathcal{L}_{ID}$, the standard Triplet Loss $\mathcal{L}_{tri}$ and a text-to-image cross-entropy loss $\mathcal{L}_{i2tce}$, which essentially computes the cross-entropy loss for logits expressed as cosine-similarities to the prompt embeddings of all the identities in the training set.

One major of the short coming of CLIP-ReID is that it optimizes prompts specific to each identity in the training set only, which cannot be generalised for real world application. Furthermore, for evaluation CLIP-ReID uses the conventional approach of nearest image search in the gallery set, which we also address with Prompt-ReID.

Re-Identification and Domain Generalization is a new line of investigation, highlighted by [17, 28, 34, 76, 79] that ReID models should be tested for their ability to make robust predictions on novel camera sources. They simulate this by training the model on a particular source dataset and testing it on an unseen target dataset. Some other works in this area propose a test time batch norm adaptation for better generalization [10, 58], generative modeling [7, 20, 36, 63, 72], transformer based approaches [29], query-adaptive convolutions [28], and the use of video sequences [57]. [31] proposes a graph sampler to improve the efficiency of the learning process the domain generalization capabilities. [35, 70] propose meta-learning to improve the generalization capabilities of networks.

Despite VLMs being known to generalize well not just on many downstream tasks but also across different domains within the tasks, these methods do not exploit them.

3. Method

3.1. Setup

Consider a general re-identification dataset, denoted as $\mathcal{D}$. This dataset is subdivided into three subsets called the training set, gallery set, and query set, which are defined as follows:

- **Training Set ($\mathcal{D}_{\text{Train}}$):** This subset is the foundation for model training, comprising N image-label pairs (x, y) . Here, x represents the image, and y corresponds to its identity label from the set $\mathcal{Y}_{\text{Train}}$.
- **Gallery Set ($\mathcal{G}$):** The gallery set encompasses image-label pairs featuring identities that do not overlap with those in the training set. These identities are uniquely represented as $\mathcal{Y}_{\mathcal{G}}$.
- **Query Set ($\mathcal{Q}$):** Reserved for the specific task of identity determination, the query set comprises images with identities distinct from the training set. These identities are denoted as $\mathcal{Y}_{\mathcal{Q}}$, forming a subset of $\mathcal{Y}_{\mathcal{G}}$.

3.2. Prompt learning for inference

We use CLIP as our base Vision-Language Model (VLM), which consists of a text encoder ($\mathcal{T}$) and an image encoder ($\mathcal{I}$). The image encoder could be a ViT or a CNN (ResNet), while the text encoder is a transformer creating a text embedding from sentences. The image encoder and the text encoder produce image and text embeddings in a shared representation space. Both the encoders are pre-trained using contrastive pretraining on a dataset containing image-caption pairs (x, t) , where x is an image, and t is its corresponding caption text. Consider a batch $\mathcal{B}$ of such pairs $(x, t) \sim \mathcal{B}$. The contrastive loss (bidirectional InfoNCE loss) can be expressed as a combination of two terms $\mathcal{L}_{\text{i2t}}$ and $\mathcal{L}_{\text{t2i}}$ expressed as:

$$\mathcal{L}_{\text{i2t}}(\mathcal{B}) = \sum_{(x, t) \in \mathcal{B}} -\log \left(\frac{e^{\text{sim}(\mathcal{I}(x), \mathcal{T}(t))}}{\sum_{(x', t') \in \mathcal{B}} e^{\text{sim}(\mathcal{I}(x'), \mathcal{T}(t'))}} \right) \quad (1)$$

$$\mathcal{L}_{\text{t2i}}(\mathcal{B}) = \sum_{(x, t) \in \mathcal{B}} -\log \left(\frac{e^{\text{sim}(\mathcal{I}(x), \mathcal{T}(t))}}{\sum_{(x', t') \in \mathcal{B}} e^{\text{sim}(\mathcal{I}(x'), \mathcal{T}(t'))}} \right) \quad (2)$$

The feature vectors $\mathcal{T}(t)$ and $\mathcal{I}(x)$ are assumed to be normalized. The final contrastive loss $\mathcal{L}_{\text{contr}} = \mathcal{L}_{\text{i2t}} + \mathcal{L}_{\text{t2i}}$

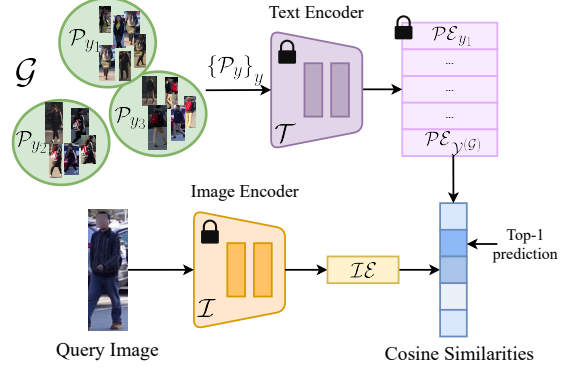


Figure 3. An overview of prompt-based inference using learnt prompts $\mathcal{P}_y$ and query image embedding.

is used for pre-training both the encoders. We shall use this objective to learn the prompts for each identity in the gallery as explained in Sec. 3.2.

For each identity $y \in \mathcal{Y}_{\mathcal{G}}$, prompt ($\mathcal{P}_y^{(\mathcal{G})}$) is introduced. Each prompt is structured as follows: $\mathcal{P}_y^{(\mathcal{G})} = [\text{A, photo, of, a, } X_{y,1}^{(\mathcal{G})}, \dots, X_{y,n}^{(\mathcal{G})}, \text{ person}]$. Here $X_{y,j}^{(\mathcal{G})}$ is a learnable token, and we will explain the process to learn it in Sec. 3.2. A prompt $\mathcal{P}_y$ when fed to the text encoder $\mathcal{T}$ gives a prompt embedding $\mathcal{PE}_y$ i.e. $\mathcal{T}(\mathcal{P}_y) = \mathcal{PE}_y$, which encodes one identity y . The prompt embedding is a 512-dimensional vector in the shared representation space of CLIP.

We learn the prompts $\mathcal{P}_y^{(\mathcal{G})}$ specifically the learnable tokens $\{X_{y,j}^{(\mathcal{G})}\}_{j=1}^n$ by optimizing the contrastive loss objective. For each image in Gallery Set, x , we set the corresponding caption to be $\mathcal{P}_y^{(\mathcal{G})}$, where $y \in \mathcal{Y}^{(\mathcal{G})}$ (Fig. 2). Using this pseudo-image-caption dataset, we minimize the contrastive loss $\mathcal{L}_{\text{cont}}$. We use SGD by batching the dataset into mini-batches of size B . Since the $\mathcal{I}$ and $\mathcal{T}$ are frozen, only the learnable tokens are updated during gradient descent. These learnable tokens $\{X_{y,j}^{(\mathcal{G})}\}_{j=1}^n$ contain fine-grained information specific to a given identity y .

During inference on the Query Set, given an image $x \sim \mathcal{Q}$, we wish to find the nearest identity in the representation space shared by the image embeddings $\mathcal{I}(x)$ and the prompt embeddings of all identities in the gallery (Fig. 3). The predicted identity $\hat{y}$ corresponds to the identity whose prompt embedding has the highest cosine-similarity. The fraction of images in $\mathcal{Q}$ for which the prediction $\hat{y} = y$ is correct, is defined as the Rank-1 accuracy.

$$\hat{y} = \arg \max_{y \in \mathcal{Y}^{(\mathcal{G})}} \frac{\langle \mathcal{I}(x), \mathcal{T}(\mathcal{P}_y) \rangle}{\|\mathcal{I}(x)\| \cdot \|\mathcal{T}(\mathcal{P}_y)\|} \quad (3)$$

4. Experiments

Architecture and pipeline: Our approach builds on CLIP-ReID. We use the image encoder and the text encoder from

Table 1. Comparison of Rank-1 accuracy of state-of-the-art CNN and ViT based re-identification methods on Market-1501, MSMT17, CUHK03, and VeRi-776 datasets. * means the accuracy was reported using higher-resolution images.

Method		Market	MSMT	CUHK	VeRi
CDNet [21]	ResNet-50	95.1	78.9	-	-
AutoReID [41]		94.5	78.2	77.9	-
PAT (2021) [26]		95.4	-	-	-
MGN [54]		96.9	-	68.0	-
CAL [44]		95.5	84.2	-	95.4
AdaSP [77]		95.5*	85.5*	84.6*	96.7
CLIP [42]		10.8	1.3	0.5	21.1
w/ Prompt-ReID	ViT	41.4	17.6	6.4	52.5
CLIP-ReID [23]		95.9	84.4	82.4	96.7
w/ Prompt-ReID		96.9	86.7	82.1	97.3
TransReID [12]	ViT	95.2*	86.2*	-	97.1*
DCAL [78]		94.7	83.1	-	96.9
CLIP [42]		12.4	4.3	1.1	31.8
w/ Prompt-ReID		39.0	24.3	5.3	62.6
CLIP-ReID [23]		95.0	88.4	80.9	97.0
w/ Prompt-ReID		96.6	91.1	83.0	98.1

CLIP as our image and text feature extractor. We use two different backbones for the image encoder: a vision transformer and a CNN. We use the ViT-B/16 as the vision transformer and ResNet50 as the CNN backbone for comparison against other SotA methods. A detailed list of hyperparameters is available in the supplementary Sec. A.

Pipeline: After the fine-tuning of the CLIP image encoder in stage 2 of CLIP-ReID, for learning the prompts $\mathcal{P}_y$, we use the same training setup as described in Sec. 3.2. We keep both encoders frozen during this step.

Training Details: As discussed in Sec 3.2, our evaluation process resembles traditional image retrieval tasks. In the conventional inference approach, the model aims to retrieve matching images from the gallery set for each query image. An important consideration is that, during retrieval, the model is expected to disregard images with the same identity and camera in the gallery set. Ignoring these similar images is crucial, as the retrieval task becomes trivial if this condition is not upheld.

To ensure fair comparisons, we adopt a “leave one camera out” training strategy. For a dataset with C cameras, we divide the gallery into C subsets, each containing images from all cameras except one. We then train C different sets of prompts, i.e., one set for each subset. Consequently, after training, we obtain C distinct prompts corresponding to each unique identity. During the evaluation, for each query image with camera c , we utilize the model trained on the gallery subset that excludes images from camera c . More details are given in the supplementart Sec. D.

It is important to note that all the results presented in this paper adhere to the “leave one camera out” strategy. This strategy ensures a fair comparison with other methods

and serves as a strict lower bound on the performance of **Prompt-ReID**. Instead of merely ignoring images with the same identity and camera, **we go a step further by excluding all images from the particular camera, enhancing the stringency of our evaluation.**

4.1. Standard ReID Task

4.1.1 Datasets and Evaluation Protocols:

We evaluate our methods on three popular person re-identification datasets, i.e., Market-1501 [71], MSMT17 [61], CUHK03 [24], and one vehicle ReID dataset, i.e., VeRi-776 [33]. For CUHK03, we use the more challenging detected subset and the new protocol [24]. For evaluation, we use the cumulative matching characteristics (CMC) at Rank-1 (R1). We do not use mAP for any comparisons.

Mean Average Precision (mAP) measures the average precision (AP) across all queries and then averages these values to give a single score. The precision at a certain rank (e.g., Precision at Rank 1, Rank 5) is defined as the number of true positive matches divided by the total number of returned results up to that rank. The AP for each query is then averaged to obtain mAP. This is relevant in the scenarios where we have multiple matches in the gallery set to retrieve, which is the case for most of the previous inference methods. Still, in our case, we perform a prompt-based retrieval where an image can only have one prompt corresponding to its identity. In this scenario, even in the best case, we can only get 1 match for each query image; hence, mAP is not the right metric.

4.1.2 Comparison with State-of-the-Art Methods:

We evaluate three person ReID datasets, Market-1501, MSMT17, CUHK03, and a vehicle ReID dataset, VeRi-776. The results are presented in Tab. 1. Note that we have not used re-ranking for any of the results.

For the ViT backbone, our model can outperform all the previous state-of-the-art methods by a significant margin. We see an improvement of 1.4% on the Market-1501 dataset, 2.1% on the CUHK03 dataset, and most notably, 2.7% improvement on the largest and most complex dataset of MSMT17. For the vehicle ReID dataset VeRi, we observe a gain of 1%. Similar trends can be seen in the CNN backbone, except on the CUHK dataset. We report 2.3% gain on the MSMT dataset and 0.6% gain on the VeRi dataset, while in the case of the CUHK dataset, we observe a drop of 0.3%. We believe this is due to the extremely low camera count (dataset statics given in supplementary Tab. 8). From these results, we observe a clear correlation between the camera counts and the performance.

4.2. Domain Generalization for Re-Identification

To gauge the robustness of our approach and its ability to generalize across unseen domains, we perform various experiments in single-source and multi-source settings for domain generalization. The model is trained on one domain and tested on a different unseen domain in the single-source setting. This tests the model’s ability to learn the domain invariant features from a dataset. The multi-source setting is an extension of the single-source setting. Here, we train a model on multiple domains and test it on a different unseen domain. This setting is more useful for the in-the-wild deployment, as it tests the models’ ability to adapt to the ever-changing environments similar to a real-life scenario. We perform these experiments using the ViT backbone. The accuracies of other competing models are reported following PAT [34].

4.2.1 Datasets and Evaluation Protocols:

We conduct experiments on four-person re-identification datasets, i.e., Market-1501, MSMT17, CUHK03, and RandPerson [56]. We use the same evaluation metric and the leave one camera out strategy as used for standard ReID tasks.

4.2.2 Single-source Domain Generalization for Re-Identification:

In the single-source setting, we use the training set of a dataset to train the model and the testing set of another unseen dataset for testing. Specifically, we use the train sets from Market, MSMT, and RandPerson to train three different models and test each of them on the test sets of Market, MSMT, and CUHK.

The results are provided in Tab. 2. Our method can surpass all the previous models by a comfortable margin. We observe the most noteworthy improvement when the model is trained on the Market dataset and tested on MSMT, which is much larger and more complicated than the Market dataset. We see an improvement of over 12% on the base CLIP-ReID and about 9% on the previous SotA model. On average, we gain 10.2% over the base CLIP-ReID model and 8.8% over the previous SotA models when the source dataset is Market. When the source dataset is MSMT, on average, we gain 12.2% over the base CLIP-ReID and 8.5% over the previous SotA models.

4.2.3 Multi-source Domain Generalization for Re-Identification:

In the multi-source setting, the testing set of one of the datasets is used for testing, while the training set of all the other datasets is used for training. To be specific, we follow the same setup as described in M3L, wherein three

Table 2. Comparison of Rank-1 accuracy for Domain Generalization where the source datasets are Market-1501 and MSMT17 respectively. On average our proposed method shows an improvement of 8.8% when trained on Market and 8.5% improvement when trained on MSMT.

Method	Training	Market	MSMT	CUHK
CLIP [42]	NA	12.4	4.3	0.6
w/ Prompt-ReID		39.0	24.3	4.5
SNR [17]	Source: Market-1501	55.1	-	-
CBN [79]		91.3	25.3	-
OSNet-AIN [76]		94.2	23.5	-
QAConv [28]		-	22.6	9.9
TransMatcher [29]		-	47.3	22.2
QAConv+GS [31]		91.6	45.9	19.1
MDA [35]		-	33.5	-
PAT [34]		92.4	42.8	25.4
CLIP-ReID [23]		95.0	44.6	40.1
w/ Prompt-ReID		96.6	56.7	48.4
SNR [17]	Source: MSMT17	70.1	-	-
QAConv [28]		72.6	-	25.3
TransMatcher [29]		80.1	-	23.7
QAConv+GS [31]		79.1	79.2	20.9
MDA [35]		79.7	-	-
PAT [34]		72.2	75.9	24.2
CLIP-ReID [23]		72.6	88.4	38.8
w/ Prompt-ReID		84.9	91.1	50.9

Table 3. Comparison of Rank-1 accuracy for Domain Generalization on Multi-Source and Randperson datasets. Multi-Source dataset comprises of Market-1501, MSMT17, CUHK03, and RandPerson. The RandPerson dataset is a synthetic dataset.

Method	Training	Market	MSMT	CUHK
QACONV ₅₀ * [28]	Source: Multi	68.6	29.9	22.9
M ³ L [70]		74.5	33.0	30.7
M ³ L _{IBN} [70]		75.9	36.9	33.1
CLIP-ReID [23]		81.5	41.2	45.6
w/ Prompt-ReID		89.6	54.8	54.1
RP Baseline [56]	Source: RandPerson	55.6	20.1	13.4
CBN [79]		64.7	20.0	-
QAConv+GS [31]		76.7	45.1	18.4
PAT [34]		73.7	45.5	20.2
CLIP-ReID [23]		71.1	32.3	26.3
w/ Prompt-ReID		83.2	44.5	34.6

datasets from Market, MSMT, CUHK, and Duke are chosen for training (only the training sets from these datasets) while the fourth one is used for testing (only the test set from this dataset). Since Duke is retracted, we replace the Duke dataset with the RandPerson dataset for the experiments following PAT.

The results are provided in Tab. 3. Once again, our approach can surpass the previous SotA models. On average, we observe a gain of 10.1% over the base CLIP-ReID model, which is the previous SotA model. Along with the multi-source results, we have also provided the performance

when the source is only RandPerson to gauge this synthetic dataset’s impact. On average, our model shows an improvement of 10.7% over the base CLIP-ReID and 4.6% over the previous SotA models.

5. Ablation Study

Here, we present some qualitative analysis of our method by analysing the fine-grained semantics learned by prompts and some necessary ablation studies to explain the effectiveness of our approach. We conduct these experiments on the MSMT and Market datasets. We analyse the information encoded within the prompts, try to explain the superior performance of prompt based inference and its ability to adapt to scenarios with very few images per identity.

5.1. Interpreting Fine-Grained Semantics

We examine the fine-grained semantics captured by the learned prompts, which synthesize common information across all images for an identity. These prompts are expected to encode more accurate semantic information than individual image embeddings. To validate the informativeness and accuracy of the semantic information learned by our prompts, we employ the DeCap (Decoding Captions) method [25]. DeCap utilizes a support memory of relevant text captions, computing cross-attention weights between the text embeddings from the support memory and the given image embeddings. These weights re-weigh the text embeddings, generating an informative initial token for the decoder, which then auto-regressively generates text captions for the input embeddings.

First we create a support memory containing potential captions describing a person’s outerwear, then using the text embedding of captions in the support memory and the prompt embedding, we employ DeCap to generate text captions. This approach allows us to analyze the semantic information captured by the learned prompts by examining the generated text captions. We then compare these captions to those generated from individual image embeddings, assessing the relative informativeness and accuracy of the learned prompts. From Fig. 4, we observe that because prompts are learned by aggregating semantic information from multiple images for a particular identity, they exhibit greater robustness and accuracy in description compared to a single image. Single images are susceptible to environmental perturbations such as illumination, pose variations, occlusion, and camera resolution. Please refer to the supplementary Sec. F.2 for a more detailed explanation of this method, some more comparisons and the exact algorithm used to generate the support memory.

5.2. Relevancy Map Comparison for Learned Prompt Embedding vs. Image Embedding

To explain the superior performance of prompt-based inference compared to nearest image search, we visualize the relevancy maps for CLIP-ReID and Prompt-ReID as shown in Fig. 1. For nearest image search, we use the embedding of a gallery image with the same identity and apply cross-attention between the [CLS] token embedding of the gallery image and the patch tokens of the query image. For prompt-based inference, we use the learned prompt embedding instead of the [CLS] token embedding. This method highlights key areas in the query image by comparing them to the gallery image’s features. The prompt embedding encodes semantic information relevant to the identity, leading the image encoder to attend to more regions of the image, explaining the robustness of our method. This reduces dependency on specific image regions, which are prone to perturbations. Please refer to the supplementary Sec. F.1 for some more comparisons.

5.3. Impact of Averaging the Features

The strength of our method lies in both the averaging effect and the regularizing effect of the text encoder. Averaging the embeddings of all images sharing the same identity enhances the common and defining characteristics while suppressing uncommon, image-specific perturbations. This improves the model’s ability to focus on distinguishing features between identities. In the following experiments, we study the impact of averaging the features of our proposed method. For a dataset with N identities, we first compute the average of all image embeddings for each identity, resulting in N averaged feature vectors. During inference, instead of computing the image-to-image distance between each image in the gallery set, we compute the distance between the query image and each of the N averaged features.

We present the results for both the vanilla CLIP and the CLIP-ReID in Table 4. For the vanilla CLIP model, averaging the image features improves performance from 12.4% to 23.9% and from 4.3% to 10.2% for the Market and MSMT datasets, respectively. With our proposed method, the accuracy nearly doubles to 39.0% and 24.3%, respectively. For the CLIP-ReID with averaged features, we observe an improvement from 95.0% to 95.9% and from 87.4% to 89.6% for the Market and MSMT datasets, respectively. With Prompt-ReID on CLIP-ReID, we gain an additional 1% to 2% accuracy, reaching scores of 96.6% and 91.1% for the Market and MSMT datasets.

5.4. Prompt Learning on few-shot images

The gallery split of MSMT17 has a heavy class imbalance. There are 3060 unique identities in the gallery set; most of the identities have about 25 images, with a few head classes having over 100 images and the tail classes having

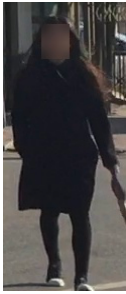
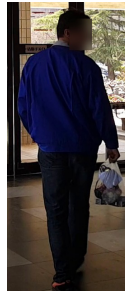
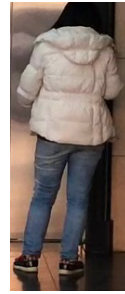
Image							
Caption generated by prompt embedding	A person wearing a brown jacket and black pants.	A person wearing a blue jacket and brown pants.	A person wearing a brown jacket and jeans.	A person wearing a plaid shirt and black pants.	A person wearing a blue shirt and a dark pants.	A person wearing a yellow jacket and white pants.	A person wearing a white jacket and gray pants.
Caption generated by image embedding	A person wearing a black jacket and a black skirt.	A person wearing a blue jacket and blue shorts.	A person wearing a blue jacket and brown pants.	A person wearing a blue plaid shirt with a plaid skirt.	A person wearing a blue shirt and a purple shirt.	A person wearing a yellow sweater and yellow pants.	A person wearing white shirt and white pants.

Figure 4. Some examples of text captions generated using the prompt and image embedding via DeCap [25]. Prompt embedding captures the semantics more accurately, while image embeddings often miss out on the exact description of attributes.

Table 4. Comparison of Rank-1 accuracy for vanilla CLIP and CLIP-ReID with, a) conventional inference, b) averaged image features for each identity, and c) Prompt-ReID

Method	Market-1501	MSMT17
CLIP [42]	12.4	4.3
w/ Avg.	23.9	10.2
w/ Prompt-ReID	39.0	24.3
CLIP-ReID [23]	95.0	87.4
w/ Avg.	95.9	89.6
w/ Prompt-ReID	96.6	91.1

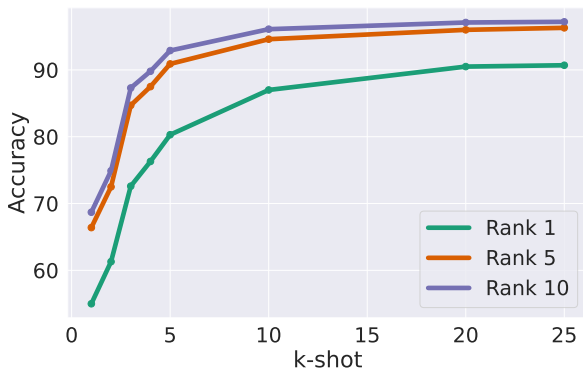


Figure 5. Results of few-shot experiments on MSMT17. $\bar{k}$ refers to the average number of images per identity.

as little as 5 images per identity. A similar trend can be observed in the train set.

In the following experiments, we wish to investigate the impact of the number of images per identity and show that our model can comfortably converge even with limited data. In our experiments, the k in k -shot refers to the average number of images per identity. We provide the results on 1 to 5, 10, 20, and 25-shot experiments in Fig. 5. For these experiments, we have scaled the total number of epochs to maintain an equal number of SGD steps, regardless of the size of the training dataset, while training. We start at an R1 accuracy of 55.0% for the 1-shot run and observe a rapid increase in performance with 61.3% R1 accuracy for the 2-shot, 72.6% for the 3-shot, 76.3% for the 4-shot, 80.3% for the 5-shot and 87.0% for the 10-shot run. When we hit the 20 images mark, the performance starts to saturate at 90.5%, which increases to 90.7% at the 25 images mark before settling at 91.1% as the full dataset performance.

6. Conclusion

In this work, we propose **Prompt-ReID**, a novel inference approach, which involves learning prompts for a gallery of identities and conducting a nearest prompt search in the shared representation space of a Vision-Language Model (VLM) for inference. It significantly outperforms the traditional method of using only an image encoder for the nearest image search. Prompts encode identity-specific semantic information, making them robust to environmental variations, unlike individual image representations. Using a text encoder during inference introduces a regularization effect, ensuring robust prompt representations even with shifts in data distribution. Another advantage of **Prompt-**

ReID is its efficiency in learning prompts from a limited number of images for each identity. The robustness of the prompts depends on the number of cameras in the surveillance network, which might limit its efficacy in settings with few cameras. An interesting avenue for further exploration could be deploying prompt learning-based inference methods in an incremental setup. We hypothesize the computational cost diminishes over time when learning a new prompt for each novel identity. The richness of prompts learned for the pre-existing identities should contribute to decreased computational overhead when learning prompts for new identities. Investigating incremental prompt learning during inference for re-identification is a promising direction for future work.

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Prompt-ReID: Prompt Based Inference for Robust Re-Identification

Supplementary Material

In this supplementary, we start with the description of all the notations used throughout the paper and a detailed list of all the hyperparameters used for the model. We then describe in details our leave one camera out training setup and provide additional experiments to study the few-shot capability of our approach, multi-source domain generalization, and impact of template used for learning the prompts. Towards the end we provide some results on explainability and interpretability of our method before going into some of the limitations of our method. We will publicly release our complete codebase and trained network weights.

This supplementary is organized as follows:

- Section A: Hyperparameters (Tab. 5 and Tab. 6)
- Section B: Notations (Tab. 7)
- Section C: Code, License, Datasets and Computation cost
 - Code and License (Sec. C.1)
 - Datasets (Sec. C.2)
 - Computation cost (Sec. C.3, Tab. 9)
- Section D: Leave one camera out setup
- Section E: Analysis on Template for the prompts (Tab. 10)
- Section F: Interpretability Analysis
 - Relevancy maps for a given prompt (Sec. F.1, Fig. 6)
 - Qualitative Results for Image and Prompt Encoding via DeCap (Sec. F.2, Fig. 7)
- Section G: Limitations
- Section H: Discussion of Ethical Issues

A. Hyperparameters

The hyperparameters for the experiments with ViT-B/16 and ResNet-50 backbones are presented in Tab. 5 and Tab. 6 respectively.

Table 5. Table of Hyperparameters for ViT-B/16 backbone.

Parameter	Market-1501	MSMT17	CUHK03	VeRi-776
Batch Size	64	64	64	64
Learning Rate	3.5e-4	3.5e-4	3.5e-4	3.5e-4
Optimizer	Adam	Adam	Adam	Adam
Weight Decay	1e-4	1e-4	1e-4	1e-4
Scheduler	Cosine	Cosine	Cosine	Cosine
Epochs	120	120	120	60
Resolution	256 × 128	256 × 128	256 × 128	256 × 256

Table 6. Table of Hyperparameters for ResNet-50 backbone.

Parameter	Market-1501	MSMT17	CUHK03	VeRi-776
Batch Size	64	64	64	64
Learning Rate	3.5e-4	3.5e-4	3.5e-4	3.5e-4
Optimizer	Adam	Adam	Adam	Adam
Weight Decay	1e-4	1e-4	1e-4	1e-4
Scheduler	Cosine	Cosine	Cosine	Cosine
Epochs	120	120	120	60
Resolution	256 × 128	256 × 128	256 × 128	256 × 256

B. Notations

We summarize all the notations used in the paper in Tab. 7.

C. Code, License, Datasets and Computation cost

C.1. Code and License

We base our work on CLIP-ReID [23]¹ in PyTorch, which is licensed under MIT License for educational purpose. We generate the English caption using the DeCap [25]² model. For the interpretability analysis we generate the relevancy maps using the Transformer-MM-Explainability [3]³ model.

C.2. Datasets

In this work we have used 5 datasets, 4 person ReID datasets, Market-1501 [71], MSMT17 [61], CUHK03 [24], RandPerson [56], and one vehicle ReID dataset, VeRi-776 [33]. The dataset statistics are presented in Tab. 8.

C.3. Computation cost

We perform all our experiments on single NVIDIA RTX A5000 GPU (24GB). The average number of SGD steps

¹<https://github.com/Syliz517/CLIP-ReID>

²<https://github.com/dhg-wei/DeCap>

³<https://github.com/hila-chefer/Transformer-MM-Explainability>

Table 7. Notation Table

Symbol	Meaning
$\mathcal{X}$	Set of images
$\mathcal{Y}$	Set of labels
x	An image
y	Label of x
$\mathcal{D}_{\text{Train}}$	Training Dataset
$\mathcal{Y}_{\text{Train}}$	Set of labels for training $\mathcal{D}_{\text{Train}}$
$\mathcal{G}$	Gallery set
$\mathcal{Q}$	Query set
$\mathcal{P}_y^{(\mathcal{G})}$	A prompt for image in $\mathcal{G}$ if label y
$X_{y,i}^{(\mathcal{G})}$	i^{th} learnable token for gallery identity y
$\mathcal{T}$	Text encoder
$\mathcal{I}$	Image encoder
$\mathcal{PE}_y$	Prompt embedding of $\mathcal{P}_y$
$\mathcal{B}$	A batch of images
$\hat{y}$	A predicted label
RK	Rank at K
$\langle A, B \rangle$	Dot product of vectors A and B
$\ \cdot\ $	Euclidean Norm
$\mathcal{L}_{\text{it}}$	Image-to-text component on InfoNCE loss
$\mathcal{L}_{\text{ti}}$	Text-to-image component on InfoNCE loss
$\mathcal{L}_{\text{ID}}$	Identity loss
V	A batch of image features
T	A batch of Text features

Table 8. ReID Dataset Statistics.

Dataset	IDs	images	cameras
Market-1501	1,501	32,217	6
MSMT17	4,101	126,441	15
CUHK03	1,467	14,097	2
RandPerson	8,000	1,801,816	-
VeRi-776	776	50,972	19

and average time for learning the prompts for inference are given in Tab. 9.

Table 9. Computation cost of Prompt-ReID.

	Market-1501	MSMT17	CUHK03	VeRi-776
SGD Steps	24,800	143,700	4,920	10,200
Average Run Time (min.)	40	120	4	8

D. Leave one camera out setup

Here we explain our leave one camera out setup with an example. As stated in the main paper, we devised this approach to ensure fair comparison with the conventional inference techniques [12]. During the conventional inference, for each query image, the model is expected to ignore images with the same identity and camera in the gallery set. Otherwise, the task of finding a match for the query image becomes trivial, since the images having the same identity and camera tend to be very similar.

While learning the prompts for our method, for a dataset with C cameras, we divide the gallery into C subsets, each containing images from all cameras except one, and then train C different sets of prompts, i.e., one set for each subset. For example, Market has 6 cameras, so we divide the gallery set into 6 subsets $\{0, 1, 2, 3, 4, 5\}$, subset 0 contains all the gallery images that are not from camera 0. We learn 6 sets of prompts for these 6 subsets. Thus we end up with 6 prompts for each identity, each learnt on a different subset. To compare with a query image with camera 0 we use the prompt with was learnt using the subset 0 for all the identities.

This strategy is much more stringent as instead of just ignoring the images with same identity and camera, we are ignoring all the images with the same camera. Thus, all the results we provide in the paper are a strict lower bound on the performance of our approach.

E. Analysis on Template for the prompts

CLIP-ReID uses the prompt “A photo of a X X X X person.”, where X means the learnable token, for all the person reid datasets. We experiment with four other prompt templates with different number of learnable, and context tokens, to study the impact of templates on model performance. We use the following templates:

- T0 : “A photo of a X X X X person.” (original)
- T1 : “A person X X X X .”
- T2 : “A photo of X person.”
- T3 : “X X X X .”
- T4 : “X .”

We perform the experiments in two different settings. In the “Only Inference” setting we use new prompt templates during our proposed inference step but use the CLIP-ReID model which was trained using the original template. In the “Throughout” setting we use a new prompt template throughout, i.e., to train the CLIP-ReID model and during the inference stage.

From the results in Tab. 10 we can conclude that the model is able to adapt to different prompt templates. The Rank 1 performance when we change the template only during the inference varies from 90.7% when the number of

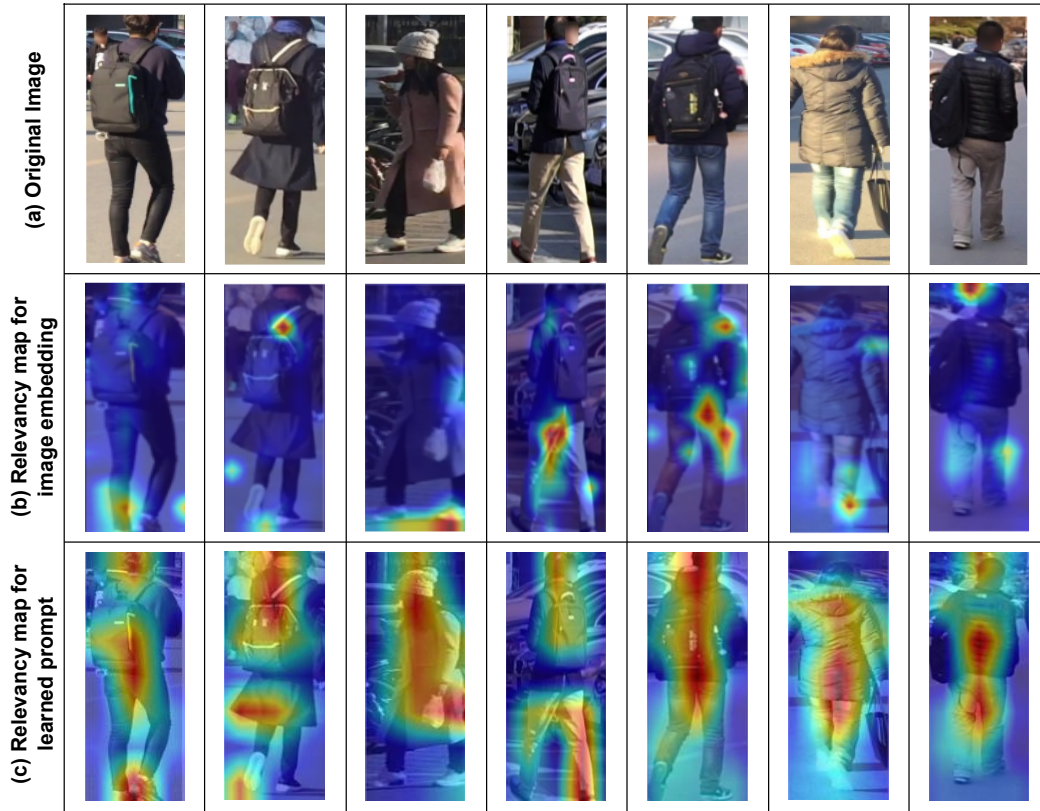


Figure 6. Some more examples to compare the relevancy map for Learned Prompt Embedding vs. Image Embedding on the MSMT17 dataset.

Table 10. Comparison of Rank-1 accuracy for ablation on Templates. The templates are as follows: T0 : “A photo of a X X X X person.” (original), T1 : “A person X X X X .”, T2 : “A photo of X person.”, T3 : “X X X X .”, T4 : “X .”

	Market-1501	MSMT17	Market-1501	MSMT17
Setting	Only Inference		Throughout	
T0	96.6	91.1	96.6	91.1
T1	96.5	91.1	96.6	91.0
T2	96.0	90.7	96.3	90.4
T3	96.3	90.8	96.2	91.0
T4	96.1	90.7	96.1	90.4

learnable tokens is 1 (T2 and T4), to 91.1% when the number of learnable tokens is 4 (T1) for the MSMT dataset. For the Market dataset we see the Rank 1 performance go from 96.0% when the number of learnable tokens is 1 (T2), to 96.5% when the number of learnable tokens is 4 (T1).

The Rank 1 performance when we use new templates for the training and inference varies from 90.4% when the number of learnable tokens is 1 (T2 and T4), to 91.0% when the number of learnable tokens is 4 (T1 and T3) for the MSMT

dataset. For the Market dataset the Rank 1 accuracy varies from 96.1% when the number of learnable tokens is 1 (T4), to 96.6% when the number of learnable tokens is 4 (T1).

F. Interpretability Analysis

F.1. Relevancy maps for a given prompt

As explained in the Sec. 5.2, we visualize the relevancy maps for CLIP-ReID and Prompt-ReID to explain the superior performance of prompt-based inference compared to nearest image search. Fig. 6 contains some more comparisons.

F.2. Qualitative Results for Image and Prompt Encoding via DeCap

We analyze the fine-grained semantics captured by the learned prompts, which aggregate information common across all images for a particular identity. Since these prompts are aggregated, we expect them to encode more accurate semantic information than individual image embeddings. To validate the informativeness and accuracy of the semantic information learned by our prompts, we employ

Image							
Caption generated by prompt embedding	A person wearing a blue jacket and brown pants.	A person wearing an orange jacket and gray pants.	A person wearing a white jacket and blue pants.	A person wearing a brown jacket and black pants.	A person wearing an orange jacket and a black pants.	A person wearing a dark jacket and white pants.	A person wearing a brown jacket and gray pants.
Caption generated by image embedding	A person wearing blue pants and a blue sweater.	A person wearing an orange jacket and orange pants.	A person wearing a white sweater and a white pants.	A person wearing brown jacket and brown pants.	A person wearing an orange jacket and red skirt.	A person wearing a white shirt and black pants.	A person wearing brown jacket and brown pants.

Image							
Caption generated by prompt embedding	A person wearing a brown jacket and black pants.	A person wearing a red sweater and a dark pants.	A woman wearing a pink coat and black pants.	A person wearing a red jacket and black pants.	A person wearing a green jacket and gray pants.	A person wearing a red jacket and black pants.	A person wearing a blue jacket and brown pants.
Caption generated by image embedding	A person wearing brown jacket and brown pants in a photo.	A person wearing a red shirt and pants is holding a red sweater.	A woman wearing a pink coat wearing a pink skirt.	A person wearing a red plaid shirt and red jacket.	A person wearing a green jacket and green pants.	A person wearing a red sweater and red pants.	A person wearing a blue shirt and blue shorts.

Figure 7. We provide more examples comparing prompts decoded from prompt embeddings and image embeddings using DeCap. Prompt embedding captions offer detailed, relevant descriptions of individuals, while image embedding captions are more generic and less informative. This demonstrates the advantage of prompt embeddings in capturing fine-grained details for person re-identification.

Algorithm 1 Generate Support Memory

```

1:  $description\_dict \leftarrow \{\dots\}$  ▷ Dictionary containing descriptions
2:  $support\_memory \leftarrow [ \ ]$  ▷ List to store the captions
3:  $Opener \leftarrow \text{"A photo of a person wearing"}$ 
4: for pieces  $\leftarrow$  cart. pdt. {upper_texture, upper_color, upper, lower, lower_color} do
5:    $text \leftarrow Opener + pieces[2] + pieces[0] + pieces[1] + \text{" and "} + pieces[4] + pieces[3] + \text{"."}$ 
6:    $support\_memory.append(text)$ 
7: end for
8: return  $support\_memory$ 

```

the DeCap (Decoding Captions) method [25]. DeCap utilizes a support memory: a corpus of text captions relevant to a specific domain. It computes cross-attention weights between the text embeddings from the support memory and

the given image embeddings. These cross-attention weights are then used to re-weight the text embeddings, generating an informative initial token for the decoder. The decoder then auto-regressively generates text captions for the input

embeddings. Alg. 1 outlines the process of generating the support memory. We create a support memory containing potential captions describing a person’s outerwear. Using the text embedding of captions in the support memory and the prompt embedding, we employ DeCap to generate text captions. This approach allows us to analyze the semantic information captured by the learned prompts by examining the generated text captions. We can then compare these captions to those generated from individual image embeddings, assessing the relative informativeness and accuracy of the learned prompts.

From Fig. 7, we observe that since prompts are learned by aggregating semantic information from multiple images for a particular identity, they exhibit greater robustness and accuracy in description compared to a single image. Single images are susceptible to environmental perturbations such as illumination, pose variations, occlusion, and camera resolution. The results in Sec. F.1 further support our findings, highlighting the effectiveness of prompt learning-based inference for ReID.

G. Limitations

One of the limitations of our approach is the strong correlation between the performance and the number of cameras. Prompt-ReID works best when it has access to images from multiple camera sources, as this helps the model learn more robust prompts that can better capture the person’s details. Among the four datasets, the most gains are observed in the case of MSMT17 and VeRi-776 with 15 and 19 cameras, respectively, while for the CUHK03 dataset with only 2 cameras, the gains are less pronounced.

H. Discussion of Ethical Issues

As with any work in the domain of ReID our PromptReID may pose several risks, for example privacy concerns among many others. In this section we wish to address these risks and propose some mitigation strategies.

First of all we ensured that all of the five datasets that we have used in this paper are valid. For the person ReID datasets, the demographic makeup does not represent the broader population but can be used for scientific experimentation. We do not claim copyright ownership of the original data, and anyone who wants to use the dataset should still be approved by the original assignee.

The following is the agreement for anyone who wants to use PromptReID:

PromptReID Agreement

This Agreement outlines the terms and conditions governing the use of PromptReID. By signing this agreement, the Recipient agrees to the following terms:

- The Recipient agrees the PromptReID does not claim Copyrights of Market-1501, CUHK03, MSMT17, RandPerson, Veri-776 and they SHOULD obtain these public datasets from these data providers.
- The Recipient agrees they must comply with ALL LICENSES of Market-1501, CUHK03, MSMT17, RandPerson, Veri-776 when they use PromptReID.
- The Recipient agrees the demographic makeup of PromptReID is not representative of the broader population.
- The Recipient agrees PromptReID should only be available for non-commercial research purposes. Any other use, in particular any use for commercial purposes, is prohibited.
- The Recipient agrees not to use the data for any unlawful, unethical, or malicious purposes.
- The Recipient agrees not to further copy, publish or distribute any portion of the PromptReID.
- The Recipient agrees [OUR INSTITUTE] reserves the right to terminate the access to the PromptReID at any time.